Towhid Ahmed

AI/ML Engineer | Machine Learning Engineer · Open to remote — US/EU time zones

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Professional Summary

Machine Learning Engineer with strong foundations in **Machine Learning**, **Deep Learning**, **MLOps**, and **applied Generative AI**. Experienced in building and deploying **end-to-end ML systems** — from classical models to **RAG pipelines** — using **FastAPI** and cloud platforms like **Render**. Strong problem-solver with **400+ DSA problems solved** and a proven ability to ship **production-grade systems** independently.

Technical Skills

Programming Languages & Data: Python (Expert), **Java (DSA)**, PostgreSQL.

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, EDA, Data Preprocessing, Feature Engineering, **Regression & Classification** Algorithms, KNN, Decision Trees, Random Forest, Naive Bayes, **TensorFlow**, **Keras**, **Scikit-learn**, **CNN**, **OpenCV** RNN/LSTM, Model Evaluation.

NLP & Generative AI: Large Language Models (LLMs), **RAG Pipelines**, Vector Databases (**Weaviate**, **FAISS**), **Cross-Encoder Re-ranking**, **LangChain**, **BERT**, Word Embeddings, **TF-IDF**.

MLOps & Deployment: MLflow, DagsHub, **Docker**, **FastAPI**, **REST APIs**, **CI/CD**, **Render**, Uvicorn, Pytest, Alembic, **Prometheus**, **GitHub Actions**

Software Engineering: 400+ DSA problems solved, Object-Oriented Programming (OOP), System Design.

Developer Tools: Pandas, NumPy, **Next.js,** GitHub, Claude, GitHub Copilot, Cursor AI, Groq API, Hugging Face Hub

Technical Experience

End-to-End MLOps Email Spam Detection System — scikit-learn | MLflow | DagsHub | FastAPI | PostgreSQL | Alembic | Prometheus | Docker | GitHub Actions (CI/CD) | Pytest **(Demo | GitHub)**

- Built a **production-grade email spam detection system** with **automated retraining**, real-time **monitoring**, and **CI/CD** deployment on Render.
- Developed a TF-IDF + Logistic Regression classifier achieving **98.5% accuracy** and **96% F1-score** on the SMS Spam Collection dataset.
- Implemented an **MLOps pipeline** with **MLflow** and DagsHub for experiment **tracking**, **registry**, and versioning.
- Designed a f**eedback-driven** retraining system where user corrections stored in **PostgreSQL** trigger **automated model retraining.**
- Built RESTful APIs with **FastAPI** supporting single and batch predictions (up to 50 inputs) with **<5ms latency**; integrated **Prometheus** for metrics monitoring and Sentry for error tracking.
- Containerized with **Docker** and deployed via **GitHub Actions CI/CD pipeline** (lint, test, build, deploy) on Render.

Potato Disease Classification — Python | TensorFlow | Keras | CNN | OpenCV | NumPy **(Demo | GitHub)**

- Built a **CNN classifier** with **TensorFlow & Keras** achieving **97.8% test accuracy**, trained with augmentation, conv/pooling layers, Adam optimizer, and cross-entropy loss.
- Built an optimized data pipeline with resizing, batching, and **`cache()`/`prefetch()`** functions; sourced and preprocessed the Kaggle PlantDisease dataset into structured batches.
- Deployed a live **Streamlit** app on Render for real-time potato leaf disease detection.

Movie Recommendation System (ML+NLP) — Python, Scikit-learn, Pandas, NumPy, TF-IDF, Cosine Similarity, Streamlit, TMDB API **(Demo | GitHub)**

- Built a **content-based movie recommendation system** using **TF-IDF** and **cosine similarity** to analyze movie metadata (genres, keywords, cast) and deliver personalized top-10 results.
- Applied TF-IDF vectorization and cosine similarity to measure similarity between movies for accurate suggestions.
- Integrated the **TMDB API** for real-time movie data and deployed on Render with public cloud access.

Advanced RAG-Based ML Research Assistant — Weaviate | Sentence Transformers | OpenAI API | Groq | Streamlit | Render **(GitHub)**

- Built an **end-to-end RAG pipeline** for semantic search and Q&A over arXiv ML research papers using **Weaviate vector database** and **OpenAI/Groq LLMs**.
- Implemented multiple document chunking strategies (fixed-size, sentence-based, semantic) with sentence-transformer embeddings for high-quality vector indexing.
- Improved retrieval accuracy using **cross-encoder re-ranking** and evaluated system performance.
- **Deployed** the **Streamlit** app on Render with a production-ready cloud setup.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology, Bangladesh

Certificates: ● **Mathematics for ML & Generative AI: Linear Algebra, Probability, Statistics, Calculus — Udemy**

- **DSA**, **Machine Learning & Deep Learning** – CodeBasics
- **MLOps** – Udemy ● **Generative AI** – Udemy